

Computational Microelectronics

Lecture 7

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L7

Homework#7

Φ_m is 4.03 eV.

The p-type doping concentration is $1.0 \times 10^{18} \text{ cm}^{-3}$.

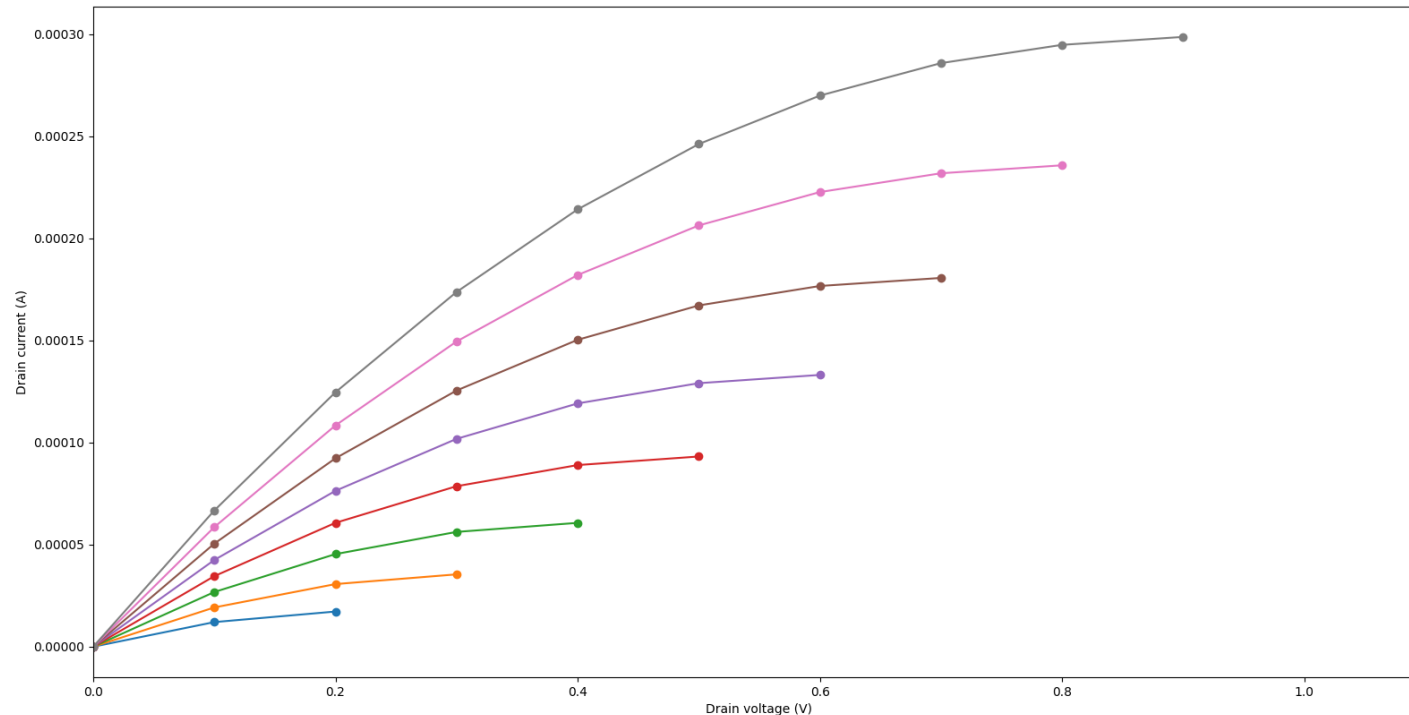
The oxide thickness is 1.2 nm.

Assume 300 K.

- Due: AM08:00, October 7

- Problem#1

- Calculate the I_D - V_D curve. ($L_G = W = 1 \text{ }\mu\text{m}$.)
- When V_D is high, it is difficult to get a converged solution...



Issues

- We have written a code to calculate I_D .
 - 1) Electron continuity equation is analytically solved.
 - 2) MOS equation and continuity equation are separately treated.
 - 3) Mobility is assumed to be constant.
 - 4) Short-channel MOSFET
 - 5) Source/drain regions
 - 6) Transient effect
 - 7) Multiple devices (Can we calculate the VTC of a CMOS inverter?)

Real space

- We avoided the real space.

- MOS equation:

$$\int \{\text{Poisson equation}\} \frac{d\phi}{dx} dx$$

- Continuity equation:

$$\int \left\{ \frac{d}{dx} \phi_n \right\} dx$$

- It was possible in special forms.

- Eventually, we must consider the real space.

- It is today's topic.

Electron continuity in the steady state

- No time derivative

$$\frac{d}{dx} J_{n,x} = 0$$

- Let us introduce $x_i = i \times \Delta x$. ($i = 0, 1, \dots, N_x - 1$)
- Integration from $x_{i-0.5}$ to $x_{i+0.5}$, (a half integer for a mid-point)

$$\int_{x_{i-0.5}}^{x_{i+0.5}} \frac{dJ_{n,x}}{dx} dx = J_{n,x}(x_{i+0.5}) - J_{n,x}(x_{i-0.5}) = 0$$

(It just states that $J_{n,x}$ is a constant.)

- How about the electron current density?

$$J_{n,x} = Q_n v_{n,x} = Q_n \mu_n \frac{d\phi_n}{dx}$$

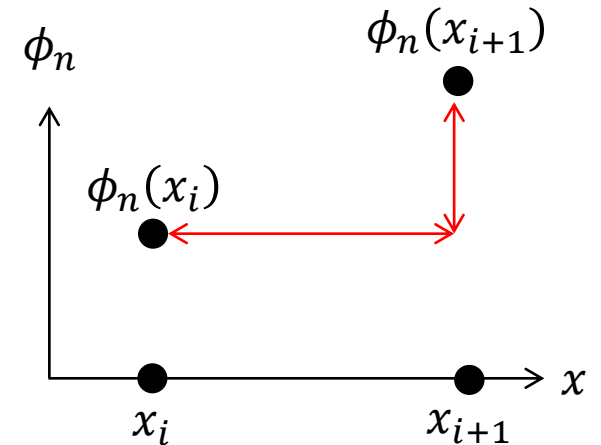
Upwind form of $J_{n,x}$

- There can be several possible forms of $J_{n,x}$.
 - We adopt the upwind form:

$$J_{n,x}(x_{i+0.5}) = Q_n(x_i) \mu_n \left. \frac{d\phi_n}{dx} \right|_{x_{i+0.5}}$$

- Linear variation of ϕ_n within a segment,

$$\left. \frac{d\phi_n}{dx} \right|_{x_{i+0.5}} \approx \frac{\phi_n(x_{i+1}) - \phi_n(x_i)}{\Delta x}$$



- By changing i ($i = 1, \dots, N_x - 2$), we can write $N_x - 2$ equations.
- Two additional equations from the boundary conditions:

$$\phi_n(x_0) = V_S \text{ and } \phi_n(x_{N_x-1}) = V_D$$

We have N_x equations.

- How can we represent them?
 - For example, $N_x = 5$ case,

$$\begin{bmatrix}
 1 & 0 & 0 & 0 & 0 \\
 \frac{Q_n(x_0)}{\Delta x} \mu_n & -\frac{Q_n(x_0) + Q_n(x_1)}{\Delta x} \mu_n & \frac{Q_n(x_1)}{\Delta x} \mu_n & 0 & 0 \\
 0 & \frac{Q_n(x_1)}{\Delta x} \mu_n & -\frac{Q_n(x_1) + Q_n(x_2)}{\Delta x} \mu_n & \frac{Q_n(x_2)}{\Delta x} \mu_n & 0 \\
 0 & 0 & \frac{Q_n(x_2)}{\Delta x} \mu_n & -\frac{Q_n(x_2) + Q_n(x_3)}{\Delta x} \mu_n & \frac{Q_n(x_3)}{\Delta x} \mu_n \\
 0 & 0 & 0 & 0 & 1
 \end{bmatrix}
 \begin{bmatrix}
 \phi_n(x_0) \\
 \phi_n(x_1) \\
 \phi_n(x_2) \\
 \phi_n(x_3) \\
 \phi_n(x_4)
 \end{bmatrix}
 =
 \begin{bmatrix}
 V_S \\
 0 \\
 0 \\
 0 \\
 V_D
 \end{bmatrix}$$

Sparse matrix

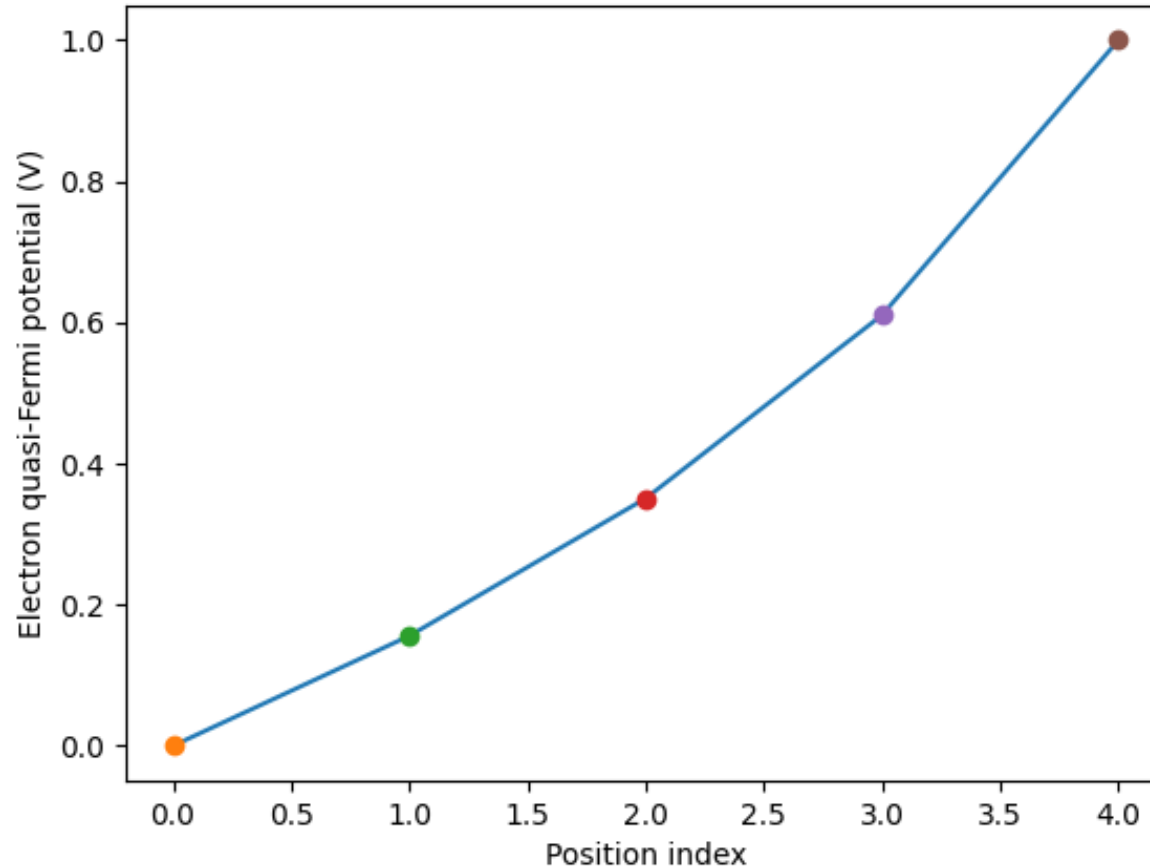
- We have $Ax = b$.
 - The matrix A contains several zeros.
 - A sparse matrix is a matrix in which most of the elements are zero.
 - Sparsity of A is $14/25 = 56\%$.
- How to construct a sparse matrix
 - C++: External libraries
 - Python: Use the `SciPy` package. (`lil_matrix` $\rightarrow$ `tocsr`)
 - MATLAB: `sparse(i, j, v, N, N);`

Sparse matrix solver

- We have $Ax = b$. (A is a sparse matrix.)
 - How can we calculate the solution vector, x ? Use a sparse matrix solver.
 - C++: External libraries
 - Python: `sparse.linalg.spsolve(A, b)`
 - MATLAB: `x = A \ b`

A simple case

- Assume that $\frac{Q_n(x_i)}{\Delta x} \mu_n = 5 - i$, $V_S = 0$, and $V_D = 1$.
 - Calculate $\phi_n(x_i)$ in two different ways.



Thank you!